

Appendix S1: Standardized Infant NeuroDevelopmental Assessment
neurological scale assessment form

Typical performance is credited with 1 point, atypical performance with a 0.

SINDA Neurological scale

Behavioural state	☐ awake, alert	☐ sleepy, tired	☐ sub-acute illness
	☐ fussy	☐ crying	
Paroxysmal events	☐ absent	☐ suspect	☐ obvious
Head	☐cm	☐ macrocephaly	☐ microcephaly
	☐ plagiocephaly	☐ atypical fontanel	☐ atypical sutures
Additional	☐ somatic symptoms	☐ sev. vis. impairment	☐ dysmorphic signs
		☐ sev. aud. impairment	☐ atypical phonation
Conclusion	☐ typical	☐ suspect	☐ atypical

Name:.....

Birth date:..... ♂ / ♀

Corr. Age:.....

Date:.....

Examiner:.....

Hospital:.....

Pos *	O,T#	item	typical (1)	atypical (0)	worst side	score
A1 Spontaneous movements (local)						
1 A	O	Head & neck & trunk	varied & symmetric	stereotyped posture, opisthotonus, asymmetric	R L	
2 A	O	Arms	varied & symmetric	stereotyped posture, asymmetric	R L	
3 A	O	Hands	varied & symmetric	stereotyped posture, asymmetric	R L	
4 A	O	Legs	varied & symmetric	stereotyped posture, asymmetric	R L	
5 A	O	Feet	varied & symmetric	stereotyped posture, asymmetric	R L	
6 Su	O	ATNR	absent, occasionally present	frequently, continuously present, asymmetric		
A2 Spontaneous movements (general)						
7 A	O	Quality	varied & symmetric & fluent & isolated movements of fingers & toes	stereotyped, jerky, jittery, startles, tremulous, sluggish, stiff, no isolated movements of fingers, toes, asymmetric	R L	
8 A	O	Quantity	moderate & changing over time	predominantly hypokinetic, predominantly hyperkinetic		
B Cranial nerves						
9 A	O	Facial appearance	varied & symmetric	expressionless, asymmetric	R L	
10 A	O	Oral motor behaviour	mouth mostly closed & tongue within the mouth & no obvious drooling	mouth mostly open, stereotyped tongue protrusion, fasciculations, obvious drooling		
11 Su, S	T	Glabella reflex	moderate threshold & moderate intensity & symmetric	low, high threshold, low, high intensity, asymmetric		
12 Su, S	O T	Eye position & eye movements	fixates & parallel position & conjugated movements	no fixation, predominant strabism, unconjugated movements, restricted motility, sunset, nystagmus	R L	
13 Su, S	T	Optical blink reflex	blinks & symmetric	absent, doubtful, delayed, asymmetric	R L	
14 Su, S	T	Pupillary reaction	direct & indirect: prompt & symmetric	absent, slow, tonic, asymmetric	R L	
15 Su, S	T	Acoustical reaction to clapping	blinks, facial reaction	absent, doubtful		
C Motor reactions to postural stimulation						
16 Su	T	Pull-to-sit	activation of neck & shoulder & arm muscles & symmetric arm activity & adequate flexion hips	head lag, active retroflexion, no, minimal muscle activation, asymmetric, inadequate hipflexion	R L	
17 P	O	Head in prone	lifts head as selective action	does not lift head, stereotyped hyperextension		
18 PS	T	Prone suspension	head in line, above trunk level	head, trunk floppy, stereotyped, opisthotonus		
19 VS	T	Vertical suspension	head upright & appropriate axillary resistance & legs varied & symmetric	poor head control, slipping through, stereotyped leg movements, asymmetric	R L	
20 VS	T	Feet touching the ground	varied feet postures & movements & symmetric	stereotyped feet postures, asymmetric	R L	
D Muscle tone						
21 Su, SS	T	Tone of neck & trunk	moderate resistance against passive movements	consistent hypotonia, consistent hypertonia, sudden changes in muscle tone		
22 Su, S	T	Resistance against passive movements, arm traction	symmetric & moderate resistance against passive movements & slight elbow flexion	consistent hypotonia, consistent hypertonia, sudden changes in muscle tone, asymmetric	R L	
23 Su, S	T	Resistance against passive movements, leg traction	symmetric & moderate resistance against passive movements & slight knee flexion	consistent hypotonia, consistent hypertonia, sudden changes in muscle tone, asymmetric	R L	
24 Su, S	T	Feet: resistance against passive movements	symmetric & moderate resistance against passive movements	consistent hypotonia, consistent hypertonia, sudden changes in muscle tone, ankle clonus, positive catch phenomenon, asymmetric	R L	
E Reflexes and reactions						
25 Su, S	T	Upper extremities: biceps reflex	symmetric positive response	areflexia, asymmetric	R L	
26 Su, S	T	Lower extremities: knee jerk & ankle clone	symmetric positive response	areflexia, tonic response, cloni, asymmetric	R L	
27 Su, S	T	Foot sole sensibility	withdrawal of legs & varied toe movement & symmetric	no reaction, stereotyped toe movements, asymmetric	R L	
28 Su, S	T	Foot sole response	varied dorsiflexion 1st toe & toe spreading & symmetric	stereotyped, tonic dorsi- or plantar flexion, no, weak response, asymmetric	R L	
SINDA Neuromotor Score						

* Pos: Position: A: All; P: Prone; PS: Prone Suspension; S: (supported) Sitting; Su: Supine; SS: Supine Suspension; VS: Vertical Suspension.
O: Observational item; T: Test item.